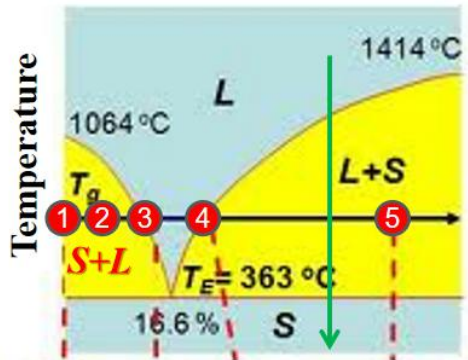
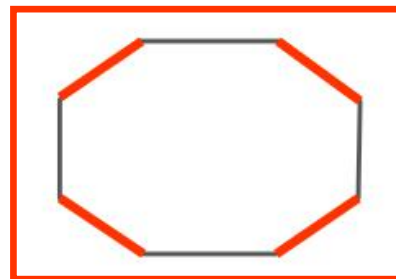
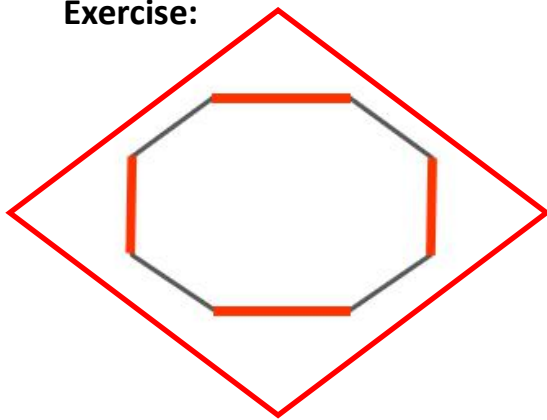


Homework: How about the progress for green line?



Firstly, the mixture of Au@Si is liquid. At a certain temperature, crystals appear. After decreasing the temperature, some crystals begin to precipitate more and liquid becomes fewer. When the temperature is lower than T_E , the whole mixture is solid.

Exercise:



Quiz2.1

(1) What are the three stages of crystal growth?

Answer:

1. The medium reaches supersaturation or supercooling;
2. Nucleation stage (homogeneous or heterogeneous nucleation);
3. Crystal growth stage.

(2) The two growth models of crystal growth and their relationships?

Answer: Layer growth theory model and Screw dislocation growth mechanism.

Relationships: Layer growth theory model is always the base of Screw dislocation growth in real cases.

(3) The internal factors that affect growth morphology of crystals.

Answer: Structure, component and defects.

(4) Except the three growth methods for single crystals, please find at least five growth methods.

Answer: CVD, PVD, Gel method, sputtering method, TGT, etc.

Exercise:

Think about the relationship between the mentioned five crystal growth models/theories. |

1. layer growth theory model
2. Screw dislocation growth mechanism
3. Bravais Law
4. Periodic bond chain (PBC) theory
5. Curie Wolfe principle (the minimum surface energy principle)

Relationship between them: all these models and mechanisms tend to minimize the surface energy.

(It's OK if you have some other logical explanations)

HOMEWORK:

1. Search papers and find at least one example for each external factor that affects crystal growth. Write a short review to explain how these external factors affect crystal growth based on these examples.

Answer is omitted here

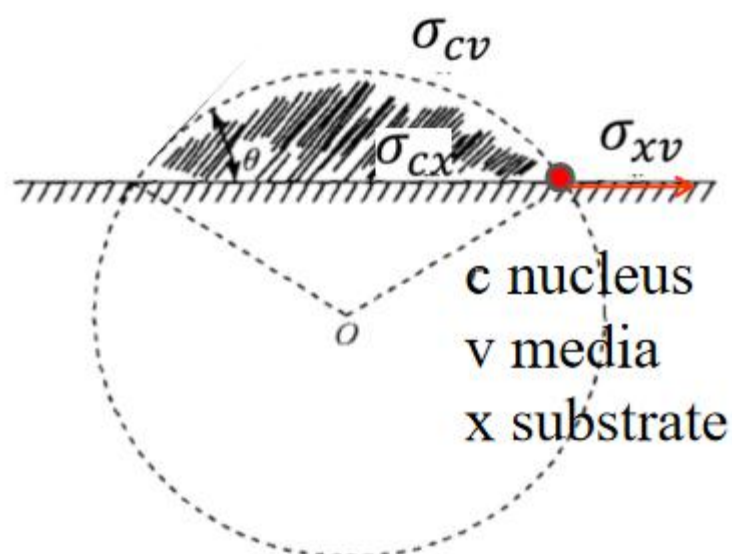
2. Find other external factors that affect crystal growth (not listed in the lecture)

Answer:

(Except temperature, impurities, viscosity, nucleation rate, vortex)

pH, stress, magnetic field, light intensity, growth order and space, etc.

Homework: drive the series of formulas in heterogeneous nucleation.



$$\Delta G_{\text{hetero}}(r) = V_c \cdot \Delta G_v + (A_{cv} \sigma_{cv} + A_{cx} \sigma_{cx} - A_{xv} \sigma_{xv})$$

$$V_c = \frac{\pi r^3}{3} (2 + \cos \theta) (1 - \cos \theta)^2$$

$$A_{cv} = 2\pi r^2 (1 - \cos \theta)$$

$$A_{cx} = \pi r^2 (1 - \cos^2 \theta)$$

$$\therefore \cos \theta = \frac{\sigma_{xv} - \sigma_{cx}}{\sigma_{cv}} \quad \Delta G_{\text{hetero}}(r) = \Delta G_{\text{homo}}(r) \cdot f(\theta)$$

$$\therefore \Delta G_{\text{hetero}}(r) = (\cos^3 \theta - 3\cos \theta + 2) \left(\frac{\pi r^3}{3} \Delta G_v + \pi r^2 \sigma_{cv} \right)$$

$$\left. \frac{d \Delta G_{\text{hetero}}(r)}{dr} \right|_{r=r^*_{\text{hetero}}} = 0$$

$$r^*_{\text{hetero}} = -2 \sigma_{cv} / \Delta G_v$$

We found that

$$\Delta G^*_{\text{hetero}}(r) = \Delta G^*_{\text{homo}}(r) \cdot f(\theta)$$

$$\therefore r^*_{\text{hetero}} = r^*_{\text{homo}}$$

$$\therefore f(\theta) \in [0, 1]$$

$$\therefore \Delta G^*_{\text{hetero}}(r) \leq \Delta G^*_{\text{homo}}(r)$$

Heterogeneous nucleation is easier than homogeneous nucleation.